

Determining the Best Routes: BGP Route Selection

1 Introduction to BGP Routes

In BGP, a **Route** is defined as a destination prefix paired with several technical **Attributes**. When a router receives multiple routes to the same prefix, it must decide which one is "best."

2 Critical BGP Attributes

Two of the most important attributes included in a BGP advertisement are:

AS-PATH and NEXT-HOP

- **AS-PATH:** A list of Autonomous Systems through which the advertisement has passed.
 - *Use 1:* To detect and prevent looping (a router rejects advertisements containing its own ASN).
 - *Use 2:* As a distance metric (shorter paths are usually preferred).
- **NEXT-HOP:** The IP address of the router interface that **begins** the AS-PATH.
 - This address is typically outside the current AS but belongs to a subnet directly attached to it.

3 Hot Potato Routing

Hot Potato routing is one of the simplest strategies used to handle outside-AS traffic.

A "Selfish" Algorithm

The Logic: Choose the route that has the **least intra-AS cost** to its NEXT-HOP router.

- **Analogy:** A packet is a "hot potato" burning your hands. You want to pass it off to another AS as quickly as possible.
- **Goal:** Minimize the transit cost within *your own* network, regardless of the total end-to-end path length.

Collaboration of Protocols

To implement hot potato routing, a router performs these steps:

1. **Learn:** Identify all possible gateways for prefix x via **BGP**.
2. **Analyze:** Consult the **Intra-AS protocol (OSPF)** to find the cost to each gateway.
3. **Select:** Choose the gateway with the smallest internal cost.
4. **Finalize:** Enter the prefix x and the corresponding local interface I into the forwarding table.

4 The BGP Route-Selection Algorithm

In real-world networks, BGP follows a more sophisticated multi-step elimination process to find the absolute best route:

The 4-Step Elimination Rules

If multiple routes to a prefix exist, BGP applies these rules in sequence:

1. **Local Preference:** Select the route with the **highest local preference** (a policy decision made by the AS administrator).
2. **Shortest AS-PATH:** From the remaining, select the route with the **fewest AS hops**.
3. **Hot Potato Routing:** From the remaining, select the route with the **closest NEXT-HOP** router (least intra-AS cost).
4. **BGP Identifiers/Tie-breakers:** If still tied, use fixed BGP identifiers as a final tie-breaker.

*Note: Unlike OSPF, which primarily optimizes for speed/cost, BGP selection is heavily driven by **Administrative Policy**.*